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Margination of platelet-sized particles in red blood cell suspensions flowing through a Y-shaped confluence microchannel

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Abstract
In blood flow through microvessels, platelets are known to be distributed in high concentrations near the vessel wall, termed 'margination' or 'near-wall excess'. At the confluence of two vessels, this preferential distribution of platelets is thought to be compromised and reconstituted in the downstream main vessel. The present study aimed to investigate the distance of this margination reconstruction from the confluence by *in vitro* experiments using platelet-sized fluorescent particles as a platelet surrogate and a Y-shaped confluence microchannel with rectangular cross sections. Fluorescence microscopy was performed using a confocal laser scanning microscope system to measure the distribution of particles in the red blood cell suspension flow. Immediately after confluence, particles were highly concentrated along a narrow band in the middle of the channel width, where particles located near the inner wall of the daughter channels flowed in. This dense band of particles faded downstream and disappeared less than 5 mm from the confluence. This margination distance is comparable to or smaller than the margination development distance in straight channels, but much smaller than that after bifurcation.

Keywords: margination, platelets, Y-shaped microchannel

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1. Introduction

In microvessels, red blood cells (RBCs) experience a lift force away from the vessel wall due to their deformability and migrate toward the vessel centre (Olla 1997, Cantat and Misbah 1999, Abkarian *et al* 2002, Geislinger *et al* 2012, Farutin and Misbah 2013, Losserand *et al* 2019, see also the of Secomb 2017). As a result, a layer depleted of RBCs (cell-free layer: CFL) is formed adjacent to the vessel wall, while the core of the vessel is rich in RBCs (RBC core) (Kim *et al* 2009). *In vivo* experimental studies of blood flow in microvessels reported high concentrations of platelets near the vessel wall, which is called 'margination' or 'near-wall excess' (Tangelder *et al* 1985, Woldhuis *et al* 1992). Since this phenomenon results from the dynamics of RBCs and their hydrodynamic interaction with platelets, not only platelets but also platelet-sized particles exhibit a concentration excess near the vessel wall, especially in the CFL, when suspended in the RBC flow (Tilles and Eckstein 1987, Eckstein *et al* 1988, Water and Eckstein 1990, Zhao *et al* 2007, Namdee *et al* 2013, D'Apolito *et al* 2015, Fitzgibbon *et al* 2015, Carboni *et al* 2016). In the RBC core, random interactions between RBCs and platelets or platelet-sized particles caused by the presence of flow shear lead to platelet or particle diffusive behaviour (shear-induced diffusion) (Eckstein *et al* 1977, Leighton and Acrivos 1987) and they are expelled toward the vessel wall (Goldsmith 1971, Turitto *et al* 1972). This results in an excess of platelets or particles in the CFL. Since platelet margination is evidently expedient in its haemostatic function, extensive studies have been conducted on this phenomenon (Crowl and Fogelson 2011, Zhao and Shaqfeh 2011, Zhao *et al* 2012, Reasor *et al* 2013, Müller *et al* 2014, Vahidkhah *et al* 2014, Fogelson and Neeves 2015, Henrique Rivera *et al* 2015, Mehrabadi *et al* 2015, 2016, 2016, Krüger 2016, Qi and Shaqfeh 2017, Chang *et al* 2018, Zavodszky *et al* 2019, Czaja *et al* 2020, Oh *et al* 2022).

One of the key quantities associated with this phenomenon is the margination distance L from the channel entrance, at which margination of platelets or platelet-sized particles has completed. Using a rectangular microchannel of height 30 μm and width 300 μm, Fitzgibbon *et al* (2015) obtained $L \sim 10$ mm for 2.15 μm diameter spherical particles in RBC suspension at a shear rate $(\dot{\gamma}) = 1000\ \mathrm{s}^{-1}$ and RBC volume concentrations $(Ht) = 0.1$ and 0.2. Carboni *et al* (2016) directly tracked spherical particles in bovine RBC suspensions at $Ht = 0.35$ flowing through a rectangular microchannel of height 40 μm and width 95 μm, at relatively low shear rates of $\dot{\gamma} = 30$–$121\ \mathrm{s}^{-1}$. From the slope of the mean square displacement with respect to time, the diffusivity of 2.11 μm diameter spherical particles was calculated to be $D \sim 15.4\ \mu\mathrm{m}^2\ \mathrm{s}^{-1}$. Employing the expression of margination distance L in terms of channel height H, average flow velocity U and diffusivity D for 2D Poiseuille flow: $L = U(H/4)^2/D$ (Zhao *et al* 2012), $D \sim 15.4\ \mu\mathrm{m}^2\ \mathrm{s}^{-1}$ implies $L \sim 7.5$ mm. Several numerical studies have also estimated the margination distance. Among them, Zhao and Shaqfeh (2011) reported $L \sim \mathrm{O}(10)$ mm for 2D channel flow of height 34 μm at $Ht = 0.2$ in the Stokes flow regime. Mehrabadi *et al* (2016) proposed a scaling law for margination distance in straight channels, in which L increases cubically with channel height and is independent of shear rate. They verified this scaling law by direct numerical simulations. Although this scaling law was for straight channels, experiments using Y-shaped bifurcating microchannels confirmed the feature that the margination distance is nearly independent of shear rate (Sugihara-Seki *et al* 2020).

Most of previous studies on platelet margination have dealt with straight channel flows. In contrast, blood vessels in the human body have a lot of bifurcations and confluences. In a previous *in vitro* study using a bifurcating microchannel, we observed an asymmetric distribution of platelet-sized particles across the channel width after bifurcation (Sugihara-Seki *et al* 2020). Particle concentrations were low near the inner wall of the daughter channel because

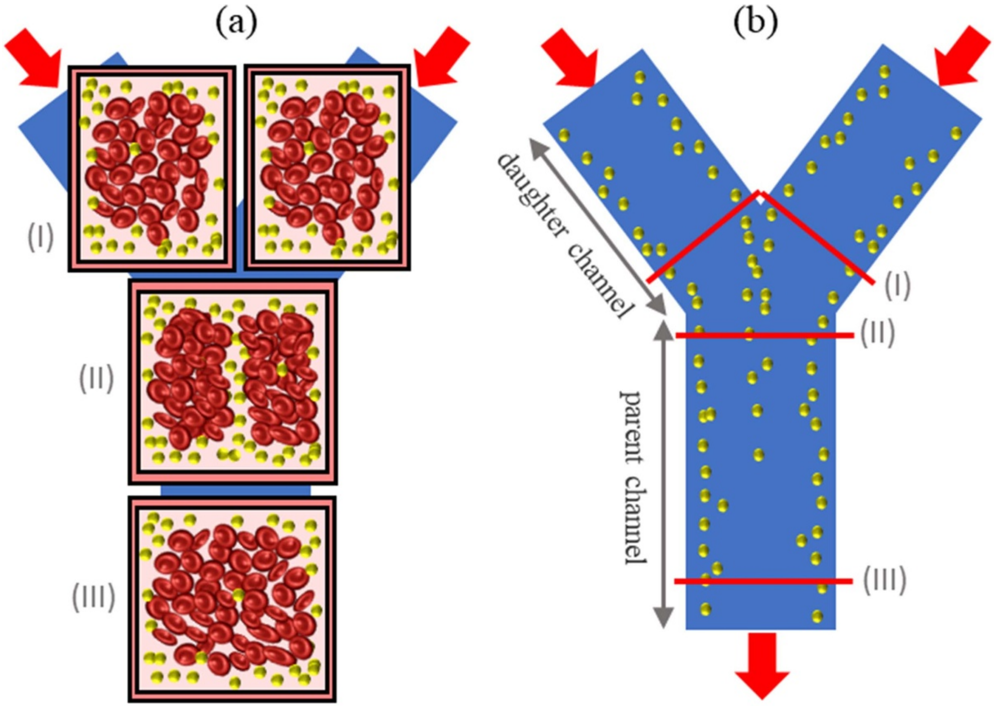

Figure 1. (a) Expected cross-sectional distributions of RBCs and particles in daughter channels (I), just behind the confluence (II), and downstream parent channel (III), and (b) particle distribution in a horizontal plane in the Y-shaped confluence channel.

the central portion of the upstream parent channel, which contained few particles due to the near-wall excess, flowed into the vicinity of the inner wall of the daughter channel. From the asymmetric distribution at the inlet of the daughter channel, the particle distribution gradually approached a symmetric one, reaching a symmetric particle concentration profile (PCP) at approximately 30 mm from the bifurcating point, i.e. $L \sim 30$ mm after bifurcation. This length is larger than the aforementioned length in straight channels. On the other hand, Bächer *et al* (2018) studied the distribution of RBCs and microparticles near a channel confluence or bifurcation, and showed by their numerical simulation that, behind the confluence of two channels, a CFL with a substantial amount of microparticles developed in the channel centre and persisted for up to 100 μm after the confluence. Since this distance corresponded to the downstream end of their computational domain, it was not known how long the particles would remain in the centre of the parent channel after confluence. This distance represents the length required to reconstruct the margination after confluence.

In the present study, to investigate the margination distance after confluence, we performed *in vitro* experiments on the margination of platelet-sized particles in the flow of RBC suspensions through a Y-shaped confluence microchannel. The upstream daughter channels and downstream parent (main) channel have rectangular cross sections. Figure 1(a) schematically illustrates the expected cross-sectional distributions of RBCs and platelet-sized particles in daughter channels (I), just behind the confluence (II), and downstream parent channel (III). If the near-wall excess of particles has developed in the daughter channels before the confluence (figure 1(I)), then a thin layer containing particles will be formed in the cross section at the confluence (figure 1(II)). These particles located in the centre of the parent channel will gradually disappear as they move downstream, resulting in a reconstruction of particle margination downstream of the parent channel (figure 1(III)). Figure 1(b) shows the expected variation in particle distribution in a horizontal plane before and after the confluence. In the present study, we measured the cross-sectional distribution of platelet-sized particles in the RBC suspension flow, and examined its variation along the main flow direction in the parent channel to determine the margination distance after confluence.

2. Material and methods

All of the procedures were performed according to the ethical policy of Kansai University. The experimental methods were described elsewhere (Sugihara-Seki *et al* 2020). Briefly, fresh human blood was sampled from young healthy volunteers and used immediately after collection. RBCs were suspended in phosphate buffered saline containing 1 wt% bovine serum albumin (Wako) and 5 wt% dextran (Dextran 40 000, Wako) at $Ht = 0.2$. The density of the medium was 1.03×10^3 kg m^{-3} and the viscosity was 2.50 mPa s at room temperature (22 °C). Spherical fluorescent polystyrene particles (Estapor F-XC300) with a mean diameter of 2.9 μm and a density of 1.05×10^3 kg m^{-3} were added to the RBC suspension at the volume concentration of $\sim$0.1%. The excitation and emission maxima of fluorescence are $\sim$470 nm and $\sim$525 nm, respectively.

Figure 2(a) illustrates the Y-shaped bifurcating flow channel made of polydimethylsiroxane (Yodaka). The bottom face was sealed with a glass coverslip for observation using a confocal laser scanning microscope system, consisting of an inverted fluorescence microscope (IX71, Olympus), a confocal scanning unit (CSU-X1, Yokogawa), a laser source (Sapphire 488, Coherent), a high-speed camera (Fastcam Mini AX50, Photron) and an image intensifier (C9016-21, Hamamatsu photonics). The upstream daughter channels have a rectangular cross section of 40 μm wide $\times$ 50 μm high and 30 mm long, and the downstream parent channel has a square cross section of 50 μm wide $\times$ 50 μm high and 10 mm long. The outlet of the parent channel was connected to a syringe, and a syringe pump (KDS270, KD Scientific) induced suspension flow at a constant flow rate of 1.0 μl min^{-1}. Assuming Newtonian fluid flow, the average wall shear rate in the parent channel is $\sim$940 s^{-1}. The Reynolds number in terms of the average flow velocity and the width of the parent channel is $\sim$0.13, indicating that inertial effects are negligible.

As shown in figures 2(b) and (c), the origin of the coordinates was set at the confluence, the x-axis was taken along the parent channel axis, the y-axis was taken across the channel width, and the z-axis was taken upward from the channel bottom. Fluorescence observations and recordings were made at six heights of $z = 2.85$–25.65 μm at $x = 0.04$, 1.0, 2.5, 5.0 and 10.0 mm (figures 2(b) and (c)). Images were recorded at a rate of 500 frames s^{-1} for 8 s (for microscope images, see figure 2 of Sugihara and Takinouchi 2022). The pixel size of the images was 0.41 μm $\times$ 0.41 μm when using a 40 $\times$ oil immersion objective (UPLSAPO40XS, Olympus). For the microscope images, the lateral (y-direction) position of each fluorescent particle was counted using an image analysis software ImageJ (NIH) to obtain a particle concentration profile (PCP), which was expressed as a histogram with a bin width of 2 μm. The histograms were symmetrized with respect to the centre ($y = 0$), considering spatial symmetry. Using the PCPs in the y-direction for various heights z and the number of particles observed, the 2D distribution of particles in the cross section (yz-plane) was obtained assuming that it is symmetric about the median plane ($z = 25$ μm).

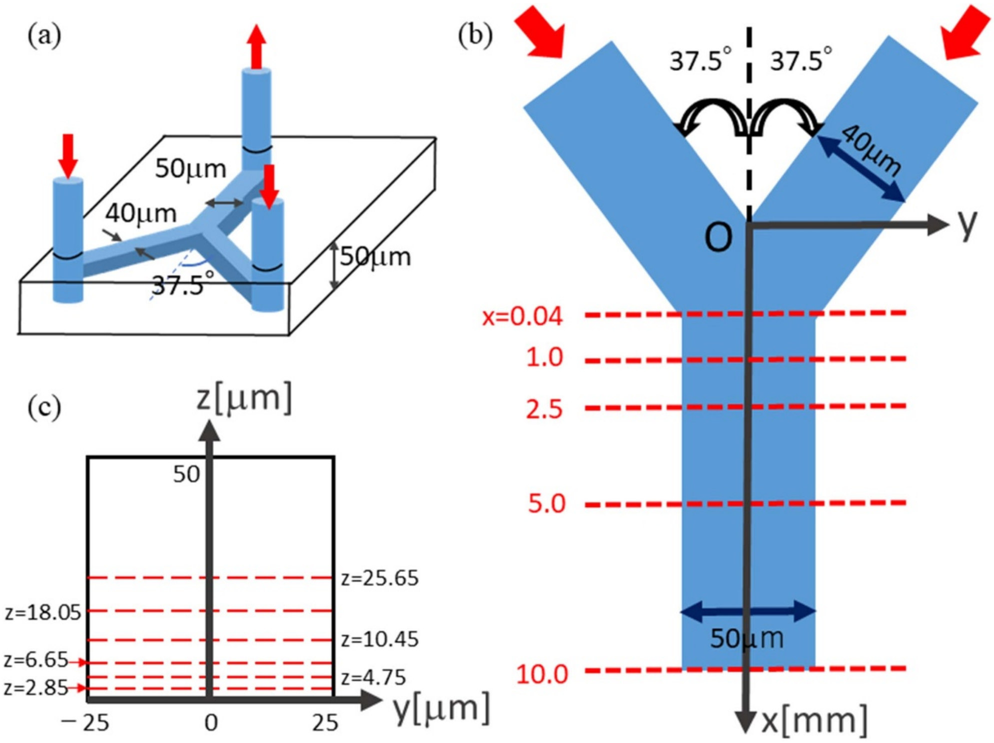

Figure 2. (a) Y-shaped confluence microchannel, (b) coordinate system and observation sites in the parent channel, and (c) observation height from the channel bottom.

3. Results and discussion

Figure 3 shows the PCPs across channel width (y-direction) at various heights z in the parent channel, at $x = 0.04$, 1.0, 2.5, 5.0, and 10.0 mm. In the most upstream cross section corresponding to figure 1(II), the PCPs shown in figure 3(a) indicate significantly higher particle concentrations near the channel centre ($y = 0$) as well as near the channel walls ($y = \pm 25\,\mu$m). As the distance (x) from the confluence increases, the particle concentration near $y = 0$ gradually decreases. In figure 3(c), the PCPs at $x = 2.5$ mm become nearly flat except near the channel walls. In further downstream corresponding to figure 1(III), the PCPs shown in figures 3(d) and (e) indicate lower particle concentrations near the centre ($y = 0$) compared to the rest at each z, except near the median plane ($z = 25\ \mu$m). At the stage of fully developed margination, the centre of the channel cross section ($y \sim 0$ and $z \sim 25\ \mu$m) was reported to have a slightly higher concentration due to low shear in this region (Oh *et al* 2022, Sugihara-Seki and Takinouchi 2022).

Using the PCPs for various z at each x shown in figure 3, we obtained the 2D distribution of particles in the cross section at $x = 0.04$, 1.0, 2.5, 5.0 and 10.0 mm (figure 4). At the most upstream cross section ($x = 0.04$ mm), particles are concentrated in a narrow band near $y = 0$ and adjacent to the channel walls (figure 1(II)). Particles near $y = 0$ gradually disperse downstream; at $x = 2.5$ mm, the band containing particles near $y = 0$ is still visible, but at $x = 5.0$ mm it is no longer visible. At $x = 5.0$ and 10.0 mm, particles are concentrated only near the channel walls, especially near the four corners of the cross section (figure 1(III)), corresponding

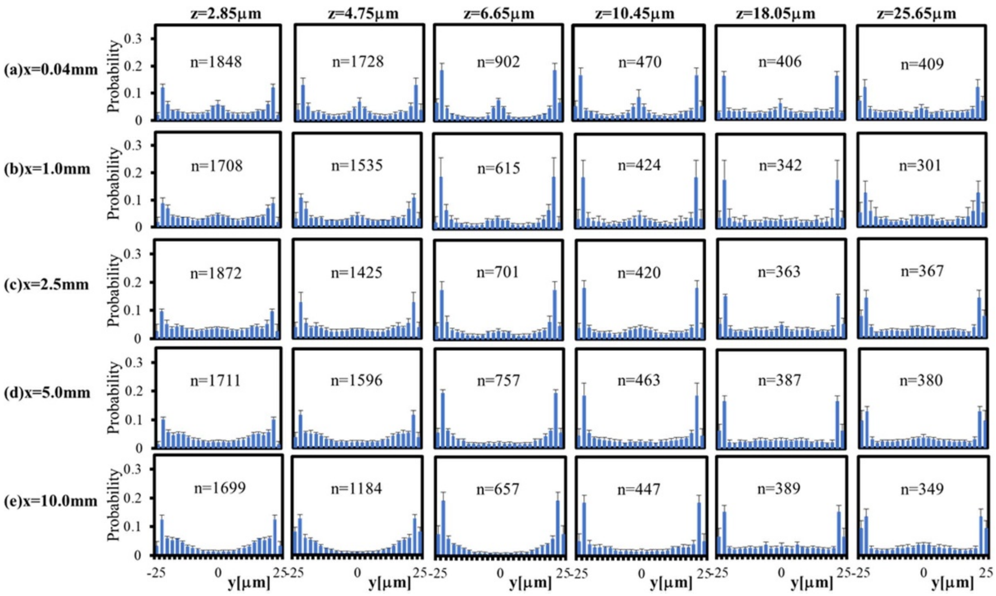

Figure 3. Particle concentration profiles (PCPs) in the parent channel. The histogram represents the average value ($N = 5$) and standard deviation. The number denotes the average number of particles detected.

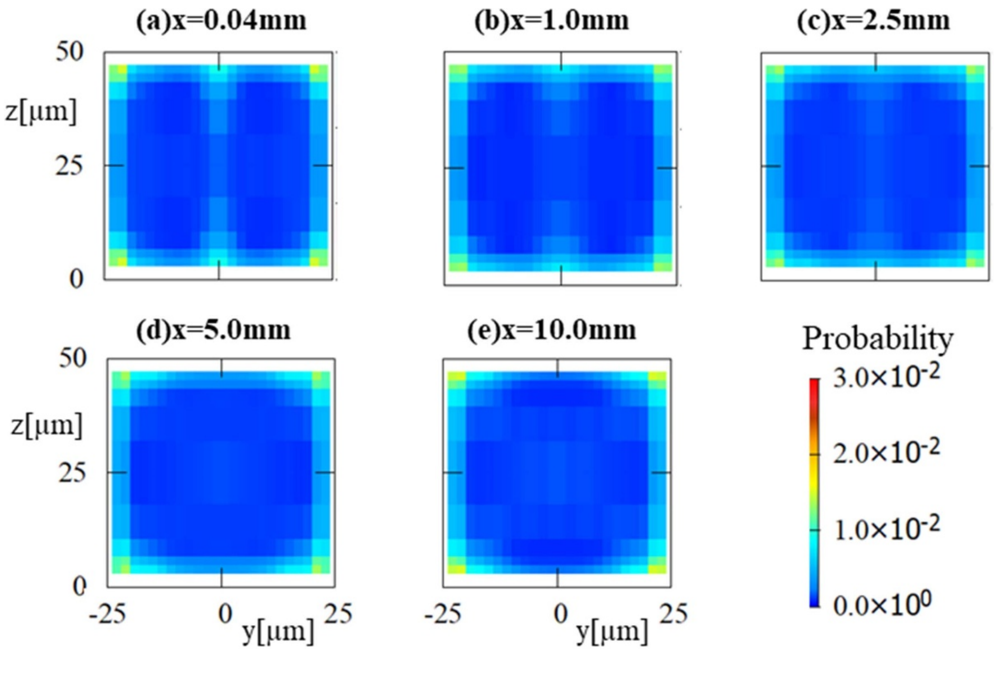

Figure 4. Cross-sectional concentration distributions of particles in the parent channel.

to the fully developed margination observed in the downstream cross section of straight rectangular channels (Sugihara-Seki and Takinouchi 2022). The present study indicated that the particle excess near $y = 0$ persisted more than 2.5 mm from the confluence and the distance, L_c, at which margination of particles fully developed after confluence was less than 5 mm. Bächer *et al* (2018) numerically studied particle distribution in RBC suspension flows through a confluence microchannel and showed that a CFL containing high concentrations of particles persists up to the downstream end of the computational domain (100 μm from confluence) and the length scale is $L_c \sim 5$ mm, agreeing with the present result.

It is not clear how the particles located in the band near $y = 0$ at the confluence spread out from that region. One possibility is that they move in the z-direction along the band toward one of the channel walls at $z = 0, H$, where $H (= 50\ \mu\text{m})$ represents the height and width of the channel. As seen in figure 1(II), a CFL is expected to develop behind the confluence near $y = 0$. Thus, in this case, the particles are most likely to move within the CFL. Another possibility is that the particles diffuse two-dimensionally toward the channel walls ($y = \pm H/2$ or $z = 0, H$) within the RBC core. For these two cases, we can estimate the diffusivity of the particles by assuming $L_c \sim 5$ mm. In the first case (1-D diffusion along the z-axis), we use the theoretical expression of the mean square displacement $\langle (\Delta z)^2 \rangle$ in time t: $\langle (\Delta z)^2 \rangle = 2D_{zz}t$. From symmetry, we consider the lower half of the channel cross section. Particles located in the band near $y = 0$ ($0 \leqslant z \leqslant H/2$) at the confluence ($x = 0$) move to the channel wall ($z = 0$) at $x = L_c$ during time $t \sim L_c/U_p$, where U_p is the particle velocity in the x-direction. Then, assuming that $U_p \sim U$ (average flow velocity) and $\langle (\Delta z)^2 \rangle \sim (H/4)^2$, the particle diffusivity is given by $D_{zz} = (H/4)^2 U_p/2L_c \sim 100\ \mu\text{m}^2\ \text{s}^{-1}$. In the second case (2-D diffusion), we assume that the particle diffusivity is isotropic in the RBC core, i.e. $D_{yy} = D_{zz} (\equiv D)$. Then, $D = \langle (\Delta y)^2 + (\Delta z)^2 \rangle U_p/4L_c \sim 100\ \mu\text{m}^2\ \text{s}^{-1}$. In both cases, similar values are obtained for the particle diffusivity after confluence.

Although $L_c \lesssim 5$ mm is comparable to or somewhat smaller than the margination distance L in straight channels, as noted in section 1, the present estimate of the particle diffusivity after confluence is larger than most of previous estimates for platelets or platelet-sized particles in the RBC core. Pioneering experimental studies by Turitto *et al* (1972) and Goldsmith and Turitto (1986) reported platelet diffusivities of $\sim$5–25 $\mu\text{m}^2\ \text{s}^{-1}$ for $Ht = 0$–0.5 and $\dot{\gamma} = 40$–$440\ \text{s}^{-1}$, and diffusivities of 2 μm microspheres of $\sim$1–20 $\mu\text{m}^2\ \text{s}^{-1}$ for $Ht = 0.2$–0.7 and $\dot{\gamma} = 5$–$20\ \text{s}^{-1}$, respectively. Recently, Carboni *et al* (2016) directly tracked 2.11 μm-diameter microspheres and obtained their diffusivity $\sim 15 \mu\text{m}^2\ \text{s}^{-1}$ for $Ht = 0.35$ and $\dot{\gamma} = 61\ \text{s}^{-1}$. 3D direct numerical simulations provided the platelet diffusivity $\sim$16 $\mu\text{m}^2\ \text{s}^{-1}$ for $Ht = 0.2$ and $\dot{\gamma} = 1000\ \text{s}^{-1}$ (Zhao *et al* 2011), $\sim$30 $\mu\text{m}^2\ \text{s}^{-1}$ for $Ht = 0.23$ and $\dot{\gamma} = 1000\ \text{s}^{-1}$ (Vahidkhaha *et al* 2014), and $\sim$27 $\mu\text{m}^2\ \text{s}^{-1}$ for $Ht = 0.2$ and $\dot{\gamma} = 503\ \text{s}^{-1}$ (Mehrabadi *et al* 2015). By a numerical simulation for a confluence channel flow, Bächer *et al* (2018) estimated the diffusivity of 3.2 μm microspheres $\sim$25 $\mu\text{m}^2\ \text{s}^{-1}$ at $Ht = 0.12$. A recent numerical study by Zavdszky *et al* (2019) reported a radial profile of platelet diffusivity in a 50 μm-diameter circular tube at $Ht = 0.35$, which increases from $\sim$70 $\mu\text{m}^2\ \text{s}^{-1}$ near the tube centre to a maximum $\sim$150 $\mu\text{m}^2\ \text{s}^{-1}$ at a radial position of 0.7 times the tube radius, followed by a sharp decrease to almost zero near the tube wall. The average value of the particle diffusivity over the cross section is not explicitly stated, but is considered to be $\sim$85 $\mu\text{m}^2\ \text{s}^{-1}$. The local shear rate at 0.7 times the tube radius is $\sim 1200\ \text{s}^{-1}$ and near the tube wall is $\sim 2200\ \text{s}^{-1}$.

The present estimate of particle diffusivity after confluence is comparatively larger than these previous values within the RBC core. This overestimation may be related to our assumption in the above estimate that $t \sim L_c/U_p \sim L_c/U$, where U_p and U represent

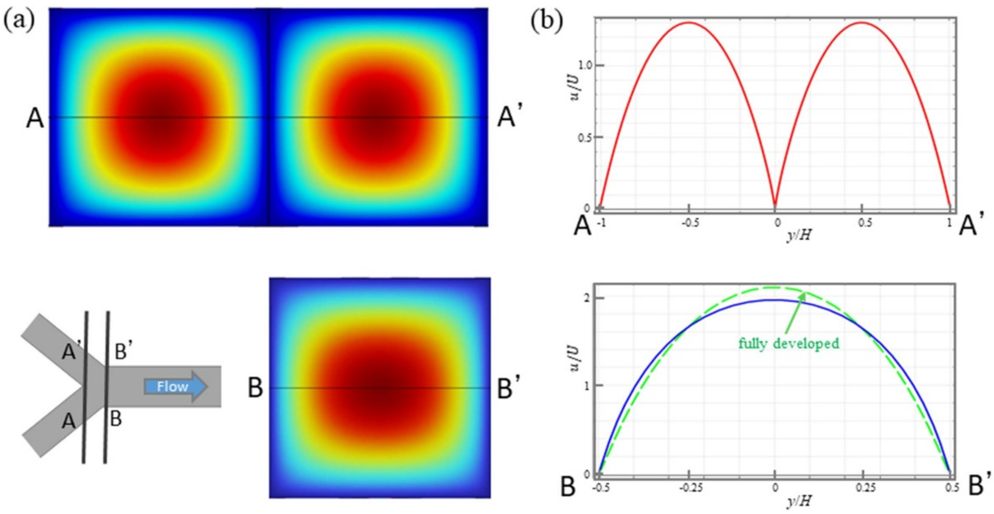

Figure 5. (a) Velocity contours and (b) velocity profiles of a Newtonian fluid, at AA′ and BB′ cross sections at $Re = 0.15$. The flow velocity was computed using COMSOL. U and H represent the average flow velocity and channel width of the parent channel, respectively.

the particle velocity and the average flow velocity of the parent channel, respectively. Figure 5 shows the velocity contours and velocity profiles at cross sections AA′ and BB′ of the Newtonian fluid flow. From the velocity contours and velocity profile at AA′, velocities of particles just behind the confluence near $y = 0$ are expected to be very small. Figure 5(b) shows that the maximum velocity at BB′ is close to that of a fully developed flow in the parent channel. Therefore, between the AA′ and BB′ cross sections, particles located near $y = 0$ are expected to be greatly accelerated to reach the fully developed flow downstream. Therefore, the particle velocity U_p after the confluence is not constant and is smaller than the value in the fully developed flow. This would increase t and, hence, the above estimate of particle diffusivity should be smaller. However, it would be important to note that, in the band containing particles near $y = 0$ at the confluence, the flow shear is very large (figure 5(b)), so that shear-induced diffusion is likely to be much higher in the band, compared to that in the RBC core in straight channels. Thus, it would be reasonable to expect that the particle diffusivity after confluence is higher than that in straight channels.

In our previous study (Sugihara-Seki *et al* 2020), the flow direction was reversed in the same Y-shaped microchannel as in the present study (figure 2(a)) to examine the particle margination after bifurcation. Figure 6(a) shows the PCPs in the daughter channel after bifurcation, measured at $z = 5.7$ μm from the bottom. The axial length from the inlet of the daughter channel is denoted as x'. The outer and inner walls of the bifurcation are located at $y' = -20$ and 20 μm, respectively. In the daughter channel immediately after the bifurcation, the particle concentration is expected to be very low near the inner wall where the flow comes from the centre of the upstream parent channel. This feature can be confirmed from figure 6(a); at the entrance of the daughter channel ($x' = 0$), the particle concentration is higher near the outer wall ($y' = -20$ μm) and lower near the inner wall ($y' = 20$ μm). As the distance from the bifurcation increases, the PCP gradually approaches symmetric one. The distance required for the PCP to become symmetric with respect to $y' = 0$ was estimated to be approximately

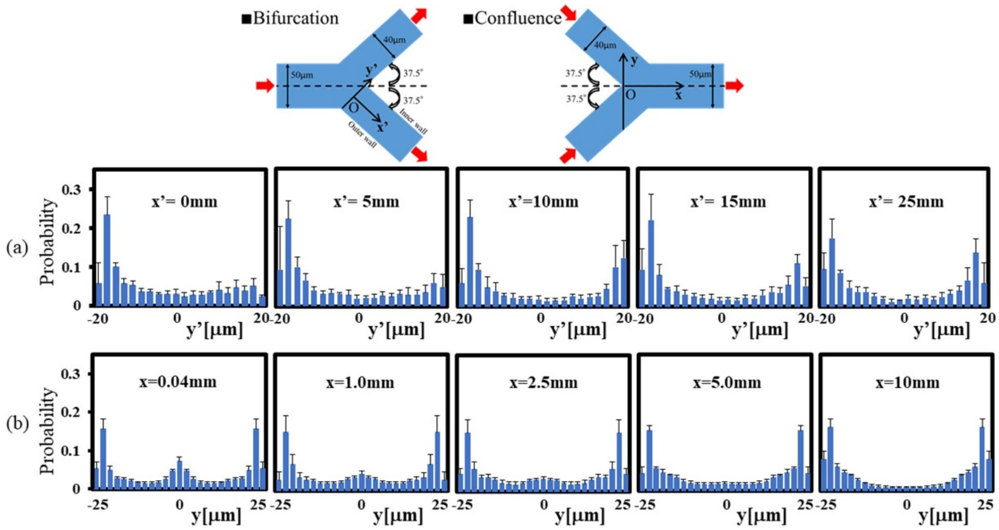

Figure 6. Particle concentration profiles (PCPs) (a) in the daughter channel after bifurcation, and (b) in the parent channel after confluence, at $z = 5.7$ μm. The flow rate was 1.0 μl min^{-1}.

30 mm, i.e. the margination distance after bifurcation $L_b \sim 30$ mm. This distance was found to be nearly independent of flow rate, consistent with a numerical study (Mehrabadi *et al* 2016).

This margination distance L_b after bifurcation is much larger than L_c after confluence obtained in the present study. For comparison, the PCPs after confluence are plotted in figure 6(b) at similar heights from the bottom, obtained by averaging the histograms for $z = 4.75$ and 6.65 μm shown in figure 3. Comparing figures 6(a) and (b), we can see that after bifurcation, particles slowly complete the full margination, in which the PCP becomes symmetric with respect to the channel centreline. One of the reasons for this may be attributed to the fact that after bifurcation, particles located mainly near the outer wall move along the bottom or top channel wall toward the inner wall to complete the margination. The direction of movement of such particles is traverse to shear. Vahidkhah *et al* (2014) showed in a numerical study that the shear-induced diffusion is smaller in the transverse direction than that along the flow shear. The slow margination feature after bifurcation can be also confirmed by a numerical study using a bifurcating channel with circular cross sections, which showed that particle depletion near the inner wall observed over approximately one-third of the circumference remained almost unchanged from immediately after bifurcation to the downstream end of the computational domain (Bächer *et al* 2018).

In the case of rectangular channels, the channel geometry may impede the full margination after bifurcation. It was found that particles in rectangular channel flows are concentrated along the channel wall, with higher concentrations near the four corners, where shear is very low (Oh *et al* 2022, Sugihara-Seki and Takinouchi 2022). Since shear-induced diffusion is very small near the corners, particles hardly emerge from that region, resulting in slower migration of particles after bifurcation. Similar results were reported in a numerical study using a Y-shaped bifurcating channel with long daughter channels (Oh 2021).

It may be important to note that after bifurcation, almost all particles are located near the channel walls, even though their distribution is not symmetric, i.e. particles are depleted along the inner wall. In other words, not fully developed in daughter channels within a distance L_b

from the bifurcation, but particle margination is achieved all along bifurcating channels. In contrast, after confluence, significant amounts of particles remain near the channel centre in the parent channel up to a distance L_c.

The margination distance $L_c \lesssim 5$ mm assessed in the present study is larger than a typical length of arterioles and venules $\sim$2 mm in the canine vasculature (Secomb 2016). Considering that the margination distance is proportional to the cube of the channel size (Mehrabadi *et al* 2016), the margination distance would be greatly reduced in smaller vessels. In this study, we employed a Y-shaped microchannel with a parent channel height and width of 50 μm. For a channel half that size, the margination distance would be one-eighth that of the present study, which is less than the above mentioned typical length of microvessels. Thus, in smaller vessels with successive confluences, platelet margination may be well developed by the next confluence. In these cases, however, certain amounts of platelets remain near the vessel centre until the margination is fully developed (by reaching the distance L_c). In addition, full margination will not be regained after bifurcation, since L_b is larger than the microvessel typical length. Thus, it is suggested that in most microvessels, fully developed margination of platelets, in which almost all platelets are distributed along the entire circumstance of the vessel wall, is rarely achieved. Platelet margination in venules may be less pronounced than in arterioles (Woldhuis *et al* 1992), and platelets may be unevenly distributed at each branching of arterioles.

4. Conclusions

In vitro experiments were conducted to investigate the 2D distribution of platelet-sized particles in the RBC suspension flow through a Y-shaped confluence microchannel with rectangular cross sections. Fluorescence microscopy measurements of the lateral distribution of particles at various heights and various distances from the confluence showed that immediately after confluence, particles were concentrated along a narrow band in the middle of the channel width, and they gradually dispersed and disappeared within 5 mm from the confluence. This margination distance after confluence is comparable to or less than the margination development distance in straight channels, but much smaller than that after bifurcation.

Acknowledgments

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